1. How each control enters the model

Each control changes exactly one parameter. Everything else stays at its calibrated value.

Control	Parameter	Formula	Symbols	Range
Personal bite protection	a (bites per mosquito per day)	$a = a_0(1 - \kappa\eta)$	κ = fraction of people who use it, η = fraction of bites it stops for a user	κ 0.2–0.6, η from AB
Larval spraying	m (adult females per person)	$m = m_0(1 - \delta)$	δ = drop in adult mosquitoes	δ 0–0.9
Wolbachia	bc (transmission per bite)	$bc \rightarrow bc(1 - p\varphi)$	p = share of mosquitoes carrying it, φ = cut in a carrier's bc	p 0–1, φ ≈ 0.9

Combined, the three multiply:

$$R_0^{control} = R_0 \bullet (1 - \kappa\eta) \bullet \sqrt{(1 - \delta)} \bullet \sqrt{(1 - p\varphi)}$$

Assumption:

- We don't split personal bite compliant and non-compliant and are just taking an average of both. This keeps the model simpler but may be less accurate when compliance is low.
- Assuming larval removal rate is proportionally similar to adult mosquito removal rate.
- Also assume that once the larval is gone, the adults will be gone too. (There is no delay)
- Wolbachia stays in the mosquito as intended all the time. Wolbachia effectiveness rate is out of scope of this model

2. Scenarios

We run four kinds of scenario, from single levers up to full packages.

Type	What we vary	Answers
Baseline	Nothing (κ = δ = p = 0)	This should be a reference for everything else
Single-lever sweep	κ 0⟶1, δ 0⟶0.95, p 0⟶1, steps of 0.05	Level needed for each lever alone; answering questions like "is larviciding alone enough?"
Pairwise grid	(κ, δ)	Answering RQ1
All three	(κ, δ) contours at p = 0, 0.5, 0.8	The balanced combination that can reduce the most cost and effort

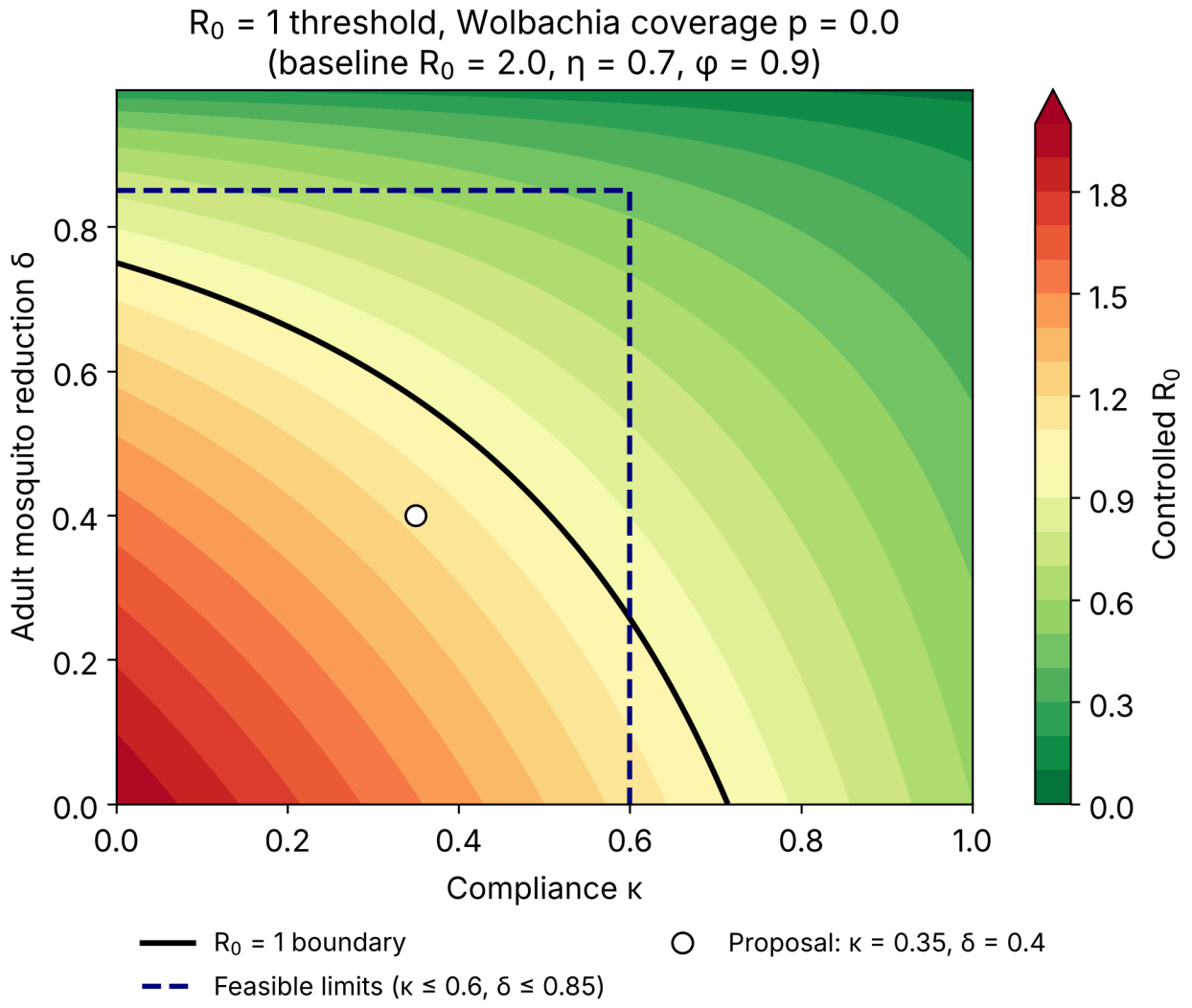
Named packages (for the 52-week case curves)

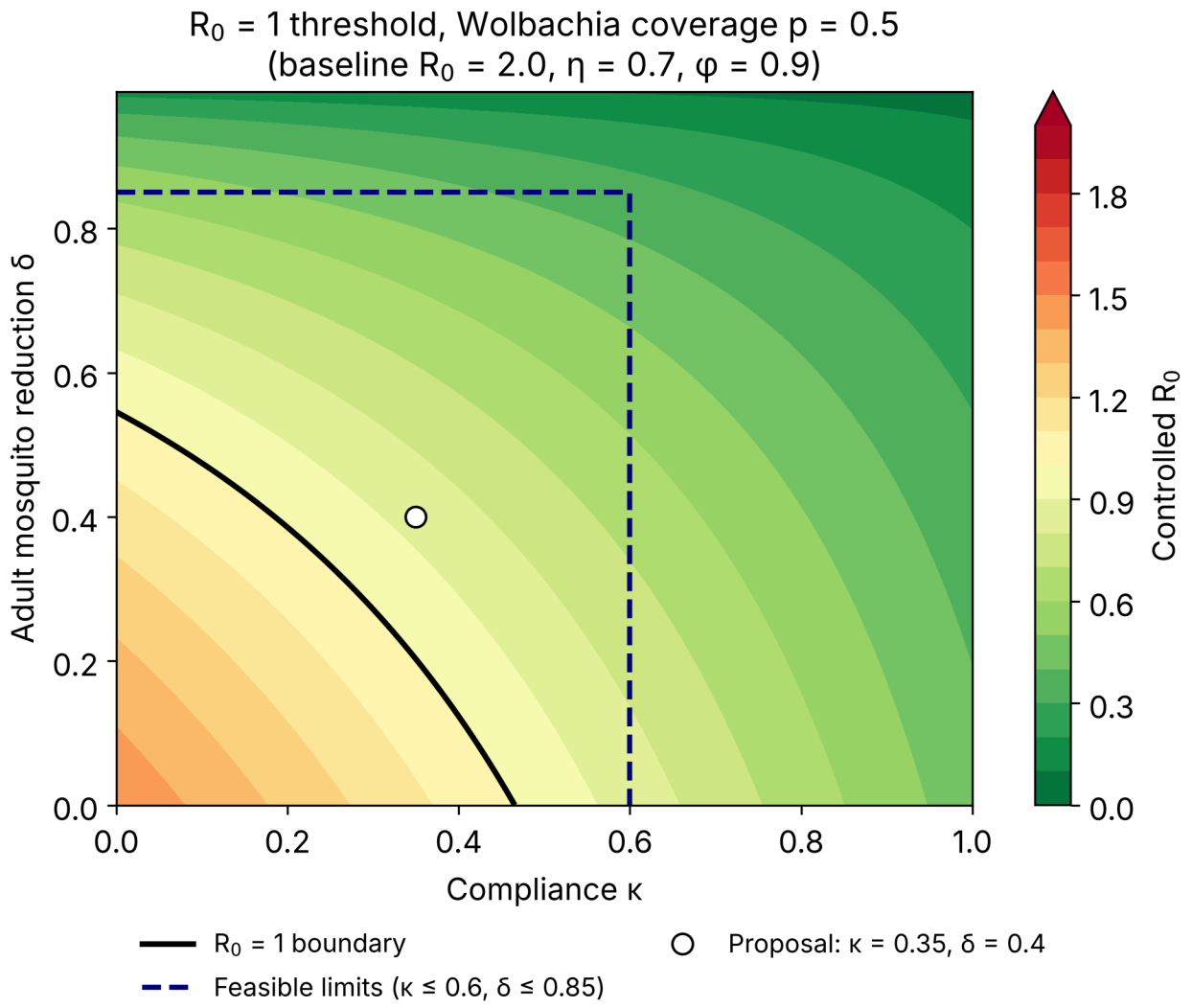
ID	Name	κ	δ	p	Why
E0	Baseline	0	0	0	Reference
E1	Realistic protection	0.35	0	0	Typical compliance
E2	Moderate spraying	0	0.4	0	Typical larval program
E3	Proposal's combination	0.35	0.4	0	In the proposal
E4	Max feasible, no *Wolbachia*	0.6	0.85	0	Without expensive new technology Wolbachia, how far can we go?
E5	Balanced package	??	??	??	Cheapest feasible point on the contour
E6	Everything at its ceiling	0.6	0.85	0.8	Upper bound

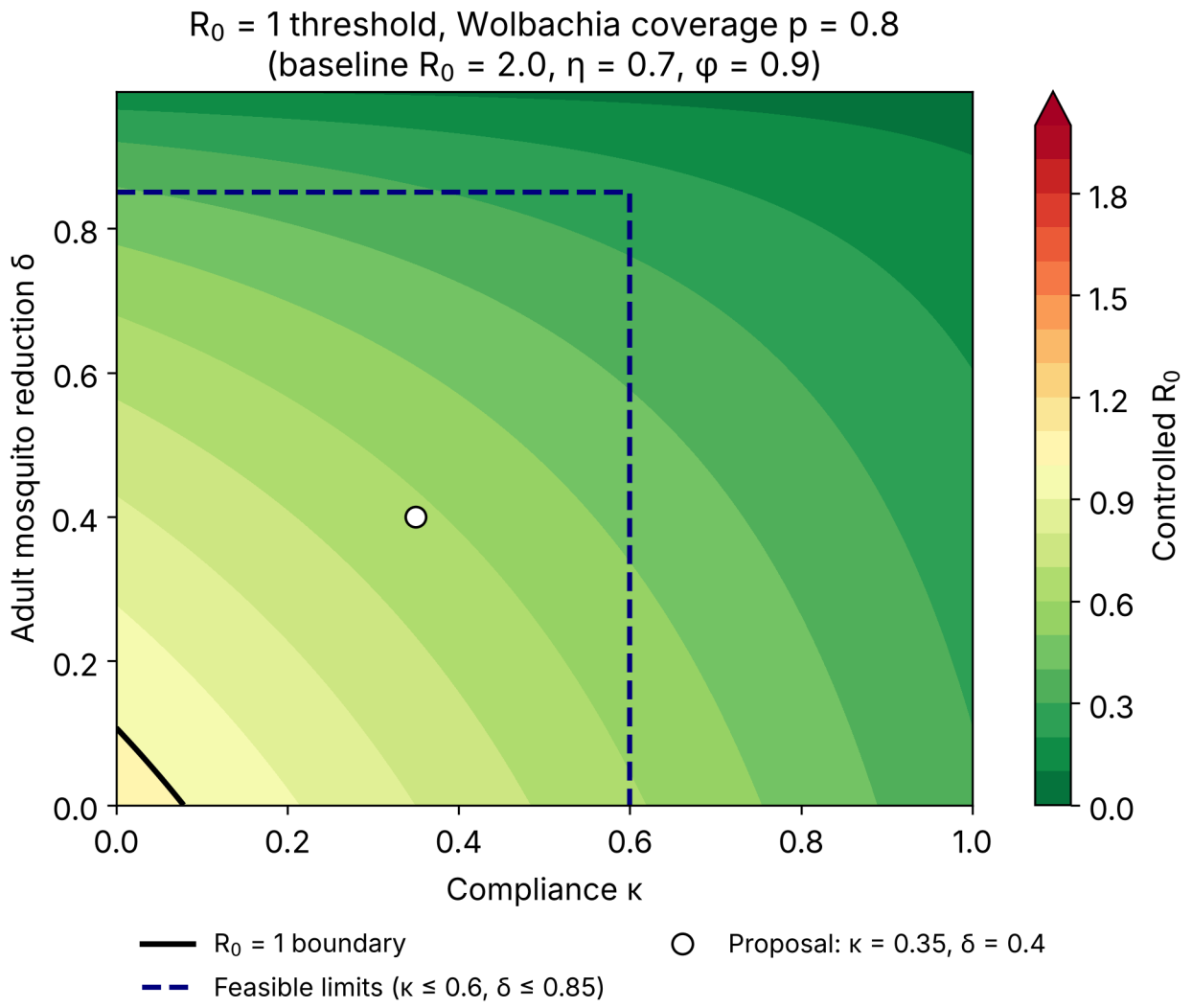
3. What we record for each scenario

Outcome	How we get it	What it tells us
Controlled R_0	Formula from Section 1	Can dengue invade a fully susceptible population with this R_0?
Effective R at season start	$R_0 \times \sqrt{(S_0/N)}$, using E1's fitted S_0	Can dengue invade a fully susceptible population with this R_0, given current immunity?
Reported cases over 52 weeks, % averted	ODE output, compared with Baseline	Total burden
Feasible?	$\kappa \leq 0.6$, $\delta \leq 0.85$, $p \leq$ attainable	Whether the package is realistic

4. Contour plot using the provisional values




R_0 = 1 threshold, Wolbachia coverage p = 0.5
(baseline R_0 = 2.0, η = 0.7, φ = 0.9)
Adult mosquito reduction δ
0.0
0.2
0.4
0.6
0.8
Compliance κ
0.0
0.2
0.4
0.6
0.8
1.0
Controlled R_0
0.0
0.3
0.6
0.9
1.2
1.5
1.8
R_0 = 1 boundary
Feasible limits (κ ≤ 0.6, δ ≤ 0.85)
Proposal: κ = 0.35, δ = 0.4


R_0 = 1 threshold, Wolbachia coverage p = 0.8
(baseline R_0 = 2.0, η = 0.7, φ = 0.9)
Adult mosquito reduction δ
Compliance κ
Controlled R_0
0.0
0.2
0.4
0.6
0.8
1.0
1.8
1.5
1.2
0.9
0.6
0.3
R_0 = 1 boundary
Feasible limits (κ ≤ 0.6, δ ≤ 0.85)
Proposal: κ = 0.35, δ = 0.4